

A very long title for my slides, which may span across two lines

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17 July 2026

The name of the conference, Where it is

The name of the session

Disclaimer...



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Manuela Joore
@ManuelaJoore

Best opening sentence [#ISPOREurope](#) from Gianluca Baio:
"statisticians should rule the world and Bayesian statisticians should rule all statisticians"



Gianluca Baio @gianlubaio · Nov 4, 2019

Ready for our session on open source models & methods!

4:52 PM · Nov 4, 2019



2



2



16



Post your reply

Reply



Ed Wilson @EdCFWilson · Nov 4, 2019
And economists should rule the Bayesian statisticians 🙄



1



1



3



Gianluca Baio @gianlubaio · Nov 4, 2019
Now don't get ahead of yourself... 😂😂



2



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Gianluca Baio
@gianlubaio

Dad to Kobi & Anna; Hubby to
[@martablangiaro](#). Also, prof of
Statistics & Health Economics [@ucl](#) &
HoD [@stats_UCL](#). Some time to spare.
mas.to/@gianlubaio

... Just so you know what you're about to get yourself into... 😊



Hello there 🖐️

This presentation will show you examples of what you can do with Quarto and **Reveal.js**, including:

- Presenting code and LaTeX equations
- Including computations in slide output
- Image, video, and iframe backgrounds
- Fancy transitions and animations
- Printing to PDF
 - Which is sometimes useful...

...and much more

Different output

- The default output is html via revealjs. This is managed by the option

```
format:  
ucl-revealjs
```

in the yml of the .qmd file.

- Alternatively, the slides can be produced in pdf directly via beamer. This is obtained by specifying

```
format:  
ucl-beamer
```

instead.

Here's a new slide without a title.

No logo

Here's a slide with no logo – just add `{ .no-logo }` to the title of the slide

Fun with colours & spaces

With a subtitle

This shows how to use colours in the slides. A bunch are pre-defined in the .css file and can be used simply as:

- The **predictive** distribution can be computed as follows.

More can be created by simply modifying the .css file to add more HEX codes.

- For example use

```
.uclblue  
color: #002855;
```

(though this colour is already defined)

Can add vertical space using fragments

```
::: style="margin-top: 40px"  
Can add vertical space using `fragments`  
:::
```

Some maths

We can also include maths, using standard LaTeX:

$$p(y \mid \mathcal{D}) = \int p(y \mid \boldsymbol{\theta})p(\boldsymbol{\theta} \mid \mathcal{D})d\boldsymbol{\theta}$$

To use colours with LaTeX, needs a little hack: `$\class{color}{expression}$`

You can define a probabilistic assumption as

$$y \mid \boldsymbol{\theta} \sim \text{Poisson}(\boldsymbol{\theta})$$

but in line it would look like this $y \sim \text{Normal}(\mu, \sigma^2)$

These use the predefined LaTeX macros, saved in the file `latex_macros.html`. More can be (re)defined by simply adding to the file, which gets pre-loaded at the beginning of the resulting `.html` output; example: `$\DIC=2345$` is typeset as $\text{DIC} = 2345$.

For instance, the command `$\dnorm$` translates to the text Normal (typeset in the current font). But can also use `$\txt{Some text}$` , which gets rendered as Some text — it's like `\text{\dots}` but with the current font

Pretty Code

(Slide 1)

- Over 20 syntax highlighting themes available
- Default theme optimized for accessibility

```
# Define a server for the Shiny app
function(input, output)

# Fill in the spot we created for a plot
output$phonePlot <- renderPlot(
# Render a barplot
)
```

Learn more: [Syntax Highlighting](#)

Multi-state/Markov models

- Assume a set S made of S “clinically relevant” states
 - Exhaustive and mutually exclusive
- The **structure** (links among nodes) describes the dynamics of disease history
 - Links connecting two states encode the assumption that a transition from the one where the link originates to the one reached by it is possible
 - Absence of a link between two states implies that the transition from one to the other is not allowed by the model

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- From one period to the next, subjects can move across the states according to the rules specified by the links
- Movements occur according to suitable **transition probabilities**

$$\pi_j = \pi_{j-1} \Lambda_j$$

where

- $\pi_j = (\pi_{1j}, \dots, \pi_{Sj})$ is the vector of probabilities for each state at time j
- $\Lambda_j = [\Lambda_{j;s',s}]$ is a transition matrix describing the probability of moving from state s to state s' at time j

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- **NB** the matrix algebra simply computes for each state s

Code Animations

(Slide 2)

- Over 20 syntax highlighting themes available
- Default theme optimized for accessibility

```
# Define a server for the Shiny app
function(input, output)

# Fill in the spot we created for a plot
output$phonePlot <- renderPlot(
# Render a barplot
barplot(WorldPhones[,input$region]*1000,
main=input$region,
ylab="Number of Telephones",
xlab="Year")
)
```

Learn more: [Code Animations](#)

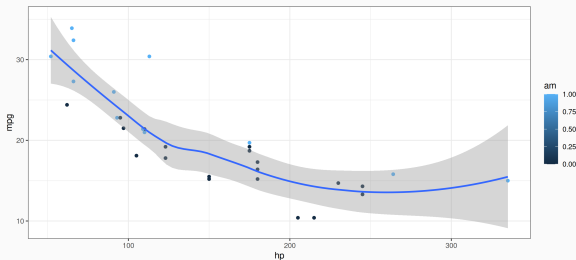
Text formatting

Blockquote

More quoted text

Executable Code

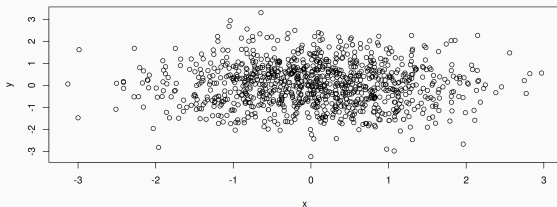
```
library(ggplot2)
ggplot(mtcars, aes(hp, mpg, color = am)) +
  geom_point() + geom_smooth(formula = y ~ x, method = "loess")
```



Learn more: [Executable Code](#)

A base plot

```
plot(rnorm(1000),rnorm(1000),xlab="x",ylab="y")
```



A section

A section

Using the tag `{.divider}` makes for some nicer css formatting (especially under `ucl-revealjs`)

Column Layout

Arrange content into columns of varying widths:

Motor Trend Car Road Tests

The data was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

	mpg	cyl	disp	hp	wt
Mazda RX4	21.0	6	160	110	2.620
Mazda RX4 Wag	21.0	6	160	110	2.875
Datsun 710	22.8	4	108	93	2.320
Hornet 4 Drive	21.4	6	258	110	3.215
Hornet Sportabout	18.7	8	360	175	3.440
Valiant	18.1	6	225	105	3.460

Learn more: [Multiple Columns](#)

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item
3. Third item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item
3. Third item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item
3. Third item

Insert pauses to make other types of content display incrementally.

Learn more: [Incremental Lists](#)

Fragments

Incremental text display and animation with fragments:

Extra fragment styling / nesting / ordering options

Fade in

Fragments

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Fade in

Slide up while fading in

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Slide left while fading in

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Highlight red (but green and blue are also built-in)

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Highlight red (but green and blue are also built-in)

Highlight custom colour (defined in scss file or can be added on the fly)

```
```css
#| echo: false
.reveal .slides section .fragment.highlight-itedared
opacity: 1;
visibility: inherit;
```